THE R-SATURATION OF A QUASI-COMPACT OPEN IS AN INTERSECTION OF QUASI-COMPACT OPENS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemmas 03HV and 0065. cited at proof lines 3852-3856. Generalization closure of the image.
- Lemma properties-lemma-quasi-compact-quasi-separated-spectral (094L). cited at proof lines 3857-3860. W' is spectral.
- Lemma topology-lemma-make-spectral-space (0A31). cited at proof lines 3860-3861. The concluding intersection representation.

Source reference. Original lemma 03HV: sources/stacks-chow/tags.txt, source lines 4581–4581. Cited at proof lines 3852-3856 to see t(s^{-1}(W)) is closed under generalization; load-bearing.

Source reference. Original lemma 0065: sources/stacks-chow/tags.txt, source lines 233–233. Cited at proof lines 3852-3856 alongside 03HV; load-bearing.

Source reference. Original lemma 094L: sources/stacks-chow/tags.txt, source lines 11554–11554. Cited at proof lines 3857-3860 to see W' is a spectral space; load-bearing.

Source reference. Original lemma 0A31: sources/stacks-chow/tags.txt, source lines 12724–12724. Cited at proof lines 3860-3861 as the concluding step; load-bearing.

Lemma 0.2. *Let S be a scheme. Let (U, R, s, t, c) be a groupoid scheme over S. Assume s and t quasi-compact and flat and U quasi-separated. Let $W \subset U$ be quasi-compact open. Then $t(s^{-1}(W))$ is an intersection of a nonempty family of quasi-compact open subsets of U.*

Proof. Note that $s^{-1}(W)$ is quasi-compact open in R. As a continuous map t maps the quasi-compact subset $s^{-1}(W)$ to a quasi-compact subset $t(s^{-1}(W))$. As t is flat and $s^{-1}(W)$ is closed under generalization, so is $t(s^{-1}(W))$, see (Morphisms, Lemma 03HV and Topology, Lemma 0065). Pick a quasi-compact open $W' \subset U$ containing $t(s^{-1}(W))$. By Properties, Lemma 094L we see that W' is a spectral space (here we use that U is quasi-separated). Then the lemma follows from Topology, Lemma 0A31 applied to $t(s^{-1}(W)) \subset W'$. □

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.